BANTRAB

Problema 2.

1. $S \to 0A0 \mid 1B1 \mid BB$
$A \to C$
$B \to S \mid A$
$C \to S \mid \epsilon$

Épsilon

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid B$
$A \to C$
$B \to S \mid A$
$C \to S$

Unario

(S,S)	$S \to B$	(S,B)	(A,A)	$A \to C$	(A,C)
(S,B)	$B \to A$	(S,A)	(A,C)	$C \to S$	(A,S)
(S,A)	$A \to C$	(S,C)	(A,S)	$S \to B$	(A,B)
(S,C)	$C \to S$	(S,S)			

(B,B)	$B \to S$	(B,S)	(C,C)	$C \to S$	(C,S)
(B,B)	$B \to A$	(B,A)			
(B,A)	$A \to C$	(B,C)	(C,S)	$S \to B$	(C,B)
			(C,B)	$B \to A$	(C,A)

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$
$A \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$
$B \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$
$C \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

BANTRAB

Inútiles / Inalcanzables

$T = \{0, 1\}$
$W_1 = \{S, A, B, C\}$
$W_2 = \{S, A, B, C\}$

$Y = \{S\}$
$Y_1 = \{S, A, B, 0, 1\}$
$Y_2 = \{S, A, B, 0, 1\}$

$S \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$
$A \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$
$B \to 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

CNF

$S \to ZAZ \mid ZZ \mid 0B0 \mid 00 \mid BB$
$A \to ZAZ \mid ZZ \mid 0B0 \mid 00 \mid BB$
$B \to ZAZ \mid ZZ \mid 0B0 \mid 00 \mid BB$

$Z \to 0$
$0 \to 1$

$C_1 \to AZ$
$C_2 \to B0$

FINAL

$S \to ZC_1 \mid ZZ \mid 0C_1 \mid 00 \mid BB$
$A \to ZC_1 \mid ZZ \mid 0C_1 \mid 00 \mid BB$
$B \to ZC_1 \mid ZZ \mid 0C_1 \mid 00 \mid BB$
$Z \to 0$
$0 \to 1$
$C_1 \to AZ$
$C_2 \to B0$

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2. $S \to aAa \mid bBb \mid \varepsilon$
$A \to C \mid a$
$B \to C \mid b$
$C \to CDE \mid \varepsilon$
$D \to A \mid B \mid ab$

Épsilon

$S \to aAa \mid aa \mid bBb \mid bb$
$A \to C \mid a$
$B \to C \mid b$
$C \to CDE \mid DE \mid CE \mid E$
$D \to A \mid B \mid ab$

Unario

(A, A)	A → C	(A, C)	(B, B)	B → C	(B, C)
(A, C)	C → E	(A, E)	(B, C)	C → E	(B, E)
(C, C)	C → E	(C, E)	(D, D)	D → A	(D, A)
			(D, A)	A → C	(D, C)
			(D, C)	C → E	(D, E)
			(D, D)	D → B	(D, B)
			(D, B)	B → C	(D, C)

$S \to aAa \mid aa \mid bBb \mid bb$
$A \to a \mid CDE \mid DE \mid CE$
$B \to b \mid CDE \mid DE \mid CE$
$C \to CDE \mid DE \mid CE$
$D \to ab \mid a \mid b \mid CDE \mid DE \mid CE$

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Inútiles / inalcanzables

$T = \{a, b\}$
$W_1 = \{S, \Delta, B, D\}$
$W_2 = \{S, \Delta, B, D\}$

$S \to a\Delta a \mid aa \mid bBb \mid bb$
$\Delta \to a$
$B \to b$
$D \to ab \mid a \mid b$

$Y = \{S\}$
$Y_1 = \{S, \Delta, B, a, b\}$
$Y_2 = \{S, \Delta, B, a, b\}$

$S \to a\Delta a \mid aa \mid bBb \mid bb$
$\Delta \to a$
$B \to b$

CNF

$S \to \Delta\Delta\Delta \mid \Delta\Delta \mid BBB \mid BB$
$\Delta \to a$
$B \to b$

$C_1 \to \Delta\Delta$
$C_2 \to BB$

FINAL

$S \to AC_1 \mid \Delta\Delta \mid BC_2 \mid BB$
$\Delta \to a$
$B \to b$
$C_1 \to \Delta\Delta$
$C_2 \to BB$

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3. $S \to ASA \mid aB \mid \varepsilon$
$A \to B \mid S$
$B \to b \mid \varepsilon$

Épsilon

$S \to ASA \mid AS \mid SA \mid S \mid aB \mid a$
$A \to B \mid S$
$B \to b$

Unarro

$(S,S)\ S \to S\ (S,S)$

$(A,A)\ A \to S\ (A,S)$

$(A,S)\ S \to ASA \mid AS \mid SA \mid aB \mid a$

$(A,A)\ A \to B\ (A,B)$

$(A,B)\ B \to b$

$S \to ASA \mid AS \mid SA \mid aB \mid a$
$A \to b \mid ASA \mid AS \mid SA \mid aB \mid a$
$B \to b$

Inútiles / inalcanzables

$T = \{a, b\}$
$W_1 = \{S, A, B\}$
$W_2 = \{S, A, B\}$

$Y = \{S\}$
$Y_1 = \{S, A, B, a\}$
$Y_2 = \{S, A, B, a, b\}$
$Y_3 = \{S, A, B, a, b\}$

BANTRAB

CNF

$S \to ASA \mid AS \mid SA \mid aB \mid a$ $\quad C_1 \to SA$
$A \to b \mid ASA \mid AS \mid SA \mid aB \mid a$
$B \to b$
$F \to a$

FINAL

$S \to AC_1 \mid AS \mid SA \mid aB \mid a$
$A \to b \mid AC_1 \mid AS \mid SA \mid aB \mid a$
$B \to b$
$F \to a$
$C_1 \to SA$

FINAL FINAL

$S \to AC_1 \mid AS \mid SA \mid FB \mid a$
$A \to b \mid AC_1 \mid AS \mid SA \mid FB \mid a$
$B \to b$
$F \to a$
$C_1 \to SA$

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